

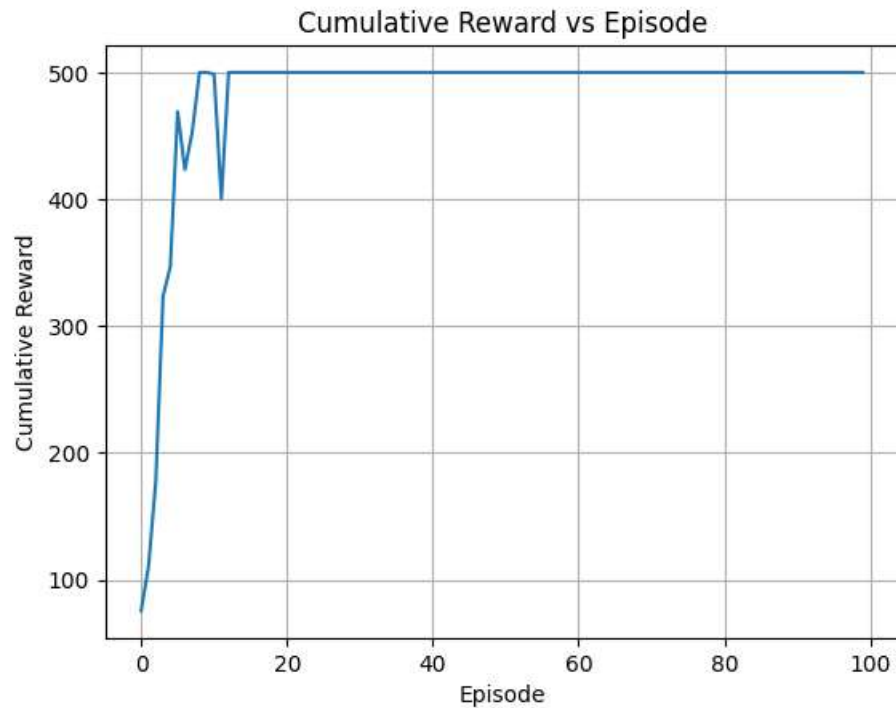
COMPARISON

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PPO:

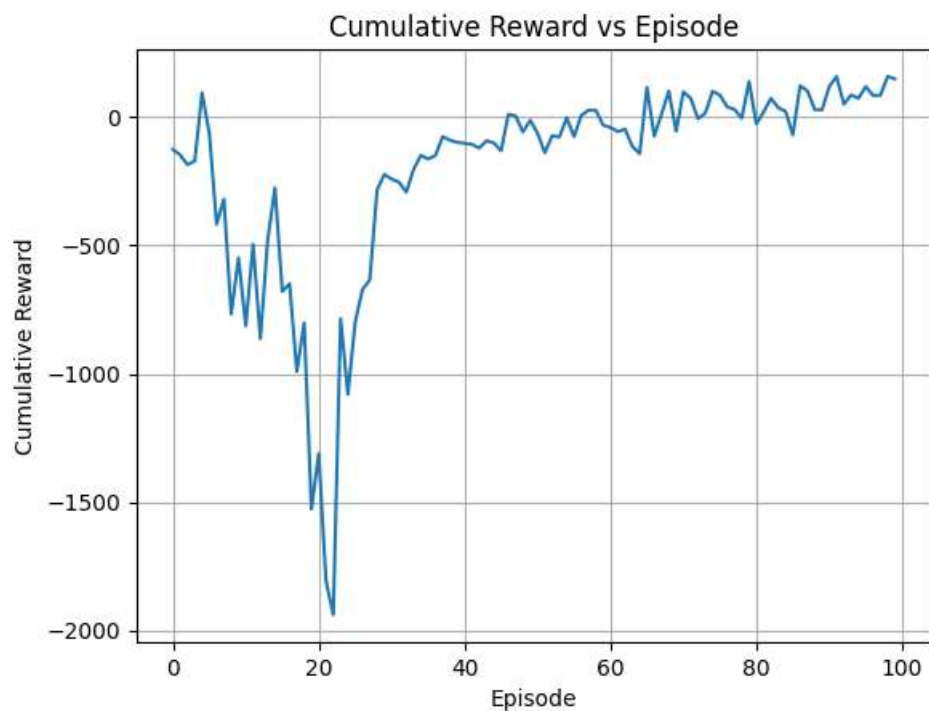
- **Cart Pole:**

1. As seen from the graph below PPO reaches a stable reward around episode 15



- **Lunar Lander:**

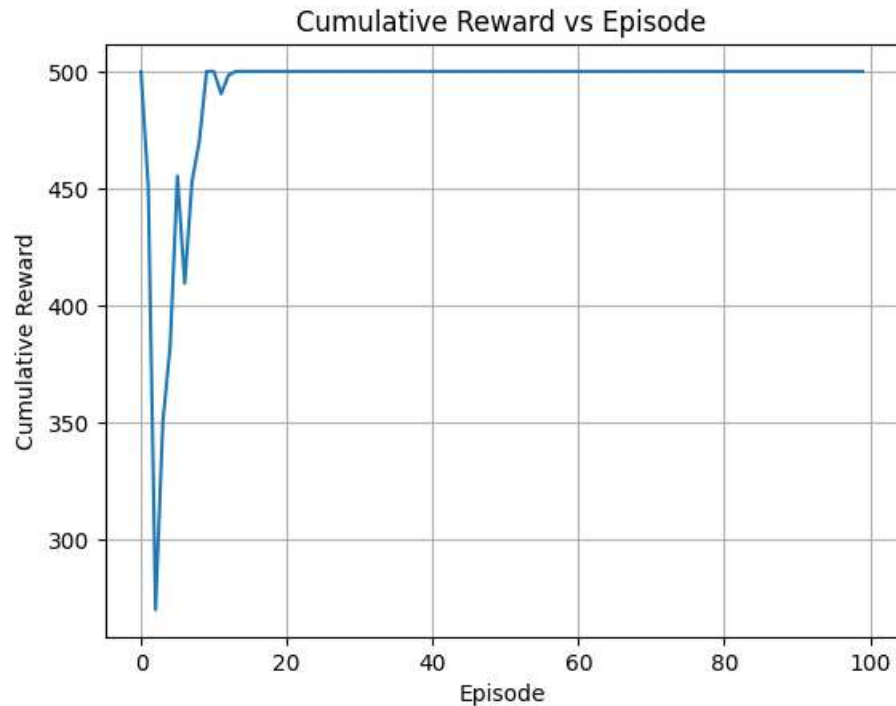
1. As seen from the graph below PPO's reward keeps fluctuating but overall it increases. Around episode 23 it starts increasing.



TRPO:

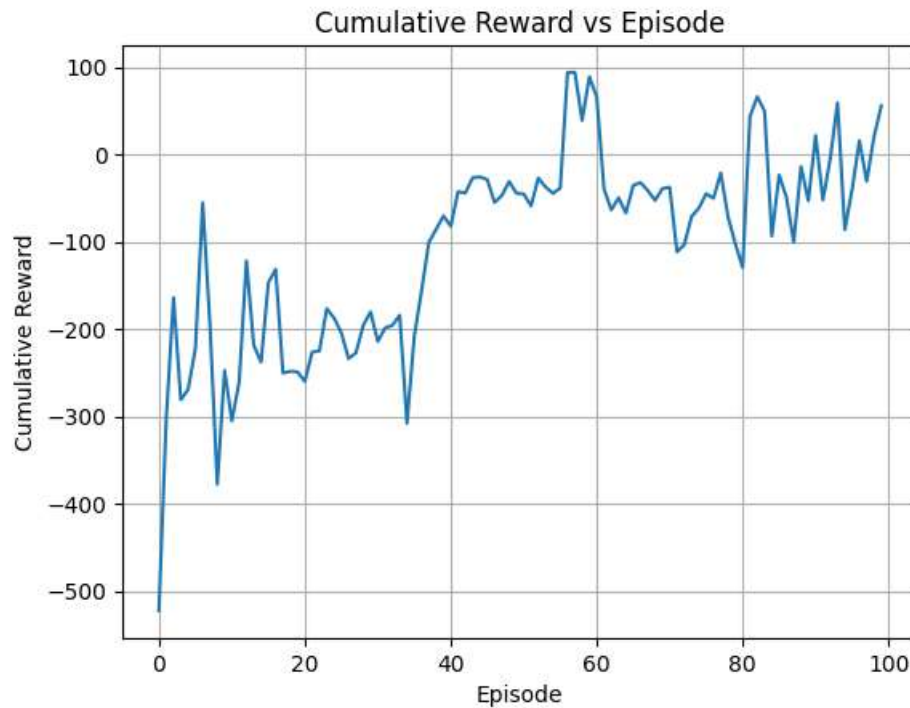
- **Cart Pole:**

1. TRPO also stabilizes around the same time as PPO for this environment



- **Lunar Lander:**

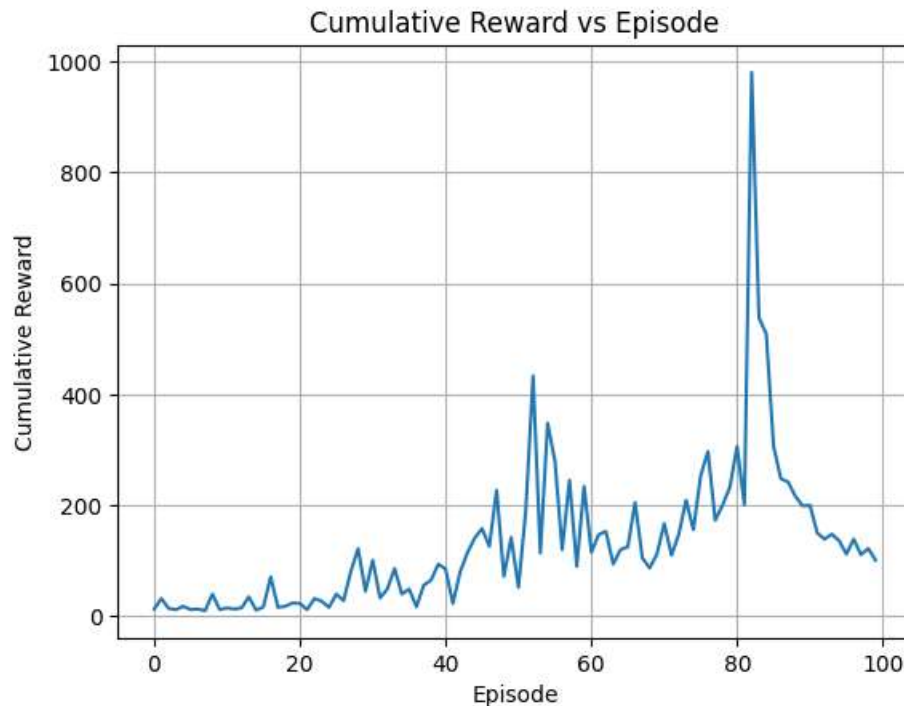
1. While TRPO does not go to such a negative value as the PPO one but the reward it gets in the end is less than PPO



REINFORCE:

- **Cart Pole:**

1. At one point this approach gives us a reward of ~1000 but then it quickly falls off and becomes worse than either PPO and TRPO



- **Lunar Lander:**

1. This approach completely fails in this approach around episode 80 it gets a positive reward. But afterwards it remains in the negatives

